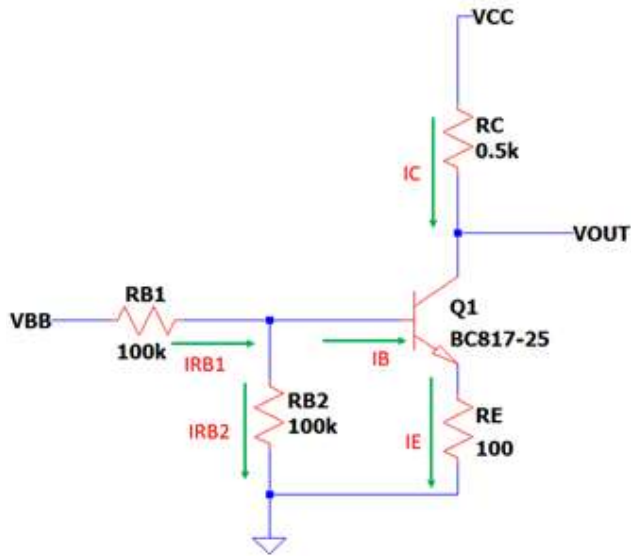




BJT Electronics How To

How to Select a Transistor

by electronicsbeliever • 0 • BJT, Electronics, How To

The transistor referred here is the bipolar junction transistor or BJT. Here is the complete guide on how to select a transistor. If you follow these, you can assure a rock solid design.

12 Key Items on How to Select a Transistor

The transistor part used in this article is from Diodes Incorporated. They are offering great options for your products. Visit them at <https://www.diodes.com/>.

1. Collector Current

From the transistor datasheet, look for the collector current rating (I_C). For instance the transistor [MJD2873Q-13](#) from Diodes Incorporated, the collector current is specified in below table as Continuous Collector Current. The

maximum limit is 2A. Thus, in your design, do not exceed the actual collector current higher to this level. Set the actual collector current to only 50% of the maximum rating and your design will be fine. You can set to higher than 50% actually but be careful and ensure your actual current calculation is accurate enough.

Absolute Maximum Ratings (@ $T_A = +25^\circ\text{C}$, unless otherwise specified.)

Characteristic	Symbol	Value	Unit
Collector-Base Voltage	V_{CB0}	70	V
Collector-Emitter Voltage	V_{CE0}	50	V
Emitter-Base Voltage	V_{EB0}	7	V
Continuous Collector Current	I_C	2	A
Peak Pulse Collector Current	I_{CM}	3	A

With a high collector current of operation, the transistor will have a shorter life that expected. Exceeding the maximum collector current will damage the transistor immediately.

2. Peak Pulse Collector Current (I_{CM})

This rating is important when the transistor is used in application in which the collector current is not straight or pure DC, for example, in switching converter, PSU and inverters. Below current waveform is a perfect example. As you can see, the current is not a pure straight line but there is a triangular component.



From the datasheet, look for the Peak Pulse Collector Current and ensure that the actual peak current of you design is not exceeding the rating. Limiting the value to 50% is still a very good idea. Higher limits will shorten the life of the transistor.

Absolute Maximum Ratings (@ $T_A = +25^\circ\text{C}$, unless otherwise specified.)

Characteristic	Symbol	Value	Unit
Collector-Base Voltage	V_{CB0}	70	V
Collector-Emitter Voltage	V_{CE0}	50	V
Emitter-Base Voltage	V_{EB0}	7	V
Continuous Collector Current	I_C	2	A
Peak Pulse Collector Current	I_{CM}	3	A

3. Collector-Emitter Voltage (V_{CE0})

The first two important ratings above on how to select a transistor are both current. Another equally important rating is the Collector-Emitter Voltage. Actually, this is the voltage seen by the transistor when the base is open. To measure this, simply get a volt meter. Put the positive probe to the collector while the negative probe to the emitter.

Absolute Maximum Ratings (@T_A = +25°C, unless otherwise specified.)

Characteristic	Symbol	Value	Unit
Collector-Base Voltage	V _{CB0}	70	V
Collector-Emitter Voltage	V _{CE0}	50	V
Emitter-Base Voltage	V _{EB0}	7	V
Continuous Collector Current	I _C	2	A
Peak Pulse Collector Current	I _{CM}	3	A

Same with the current ratings, setting to 50% limit is a good idea. However, in high voltage and high power applications, 50% limit is expensive. Why is this? For example in an inverter circuit, the actual voltage seen in the collector is 1000V (this is the maximum value). 50% limit means the transistor voltage rating must be 2000V. 2000V is not common and very expensive so making the design not practical. Instead, the next abundantly available voltage rating is 1200V so use it. This is still okay as long as you have conducted robustness and reliability testing.

For low voltage application on the other hand, there are lot of transistor options that do not increase cost, so this is not an issue.

4. Emitter-Base Voltage (V_{EB0})

This is the voltage across the emitter to base junction while the collector is open. The base-emitter of a transistor is basically a diode. In other words, the emitter-base voltage is the maximum reverse voltage that can be applied across to this diode.

Absolute Maximum Ratings (@T_A = +25°C, unless otherwise specified.)

Characteristic	Symbol	Value	Unit
Collector-Base Voltage	V _{CB0}	70	V
Collector-Emitter Voltage	V _{CE0}	50	V
Emitter-Base Voltage	V _{EB0}	7	V
Continuous Collector Current	I _C	2	A
Peak Pulse Collector Current	I _{CM}	3	A

Take note not to exceed this value. Otherwise, the transistor will get damage right away.

5. Collector-Base Voltage (V_{CBO})

This is the voltage across the collector to base junction when the emitter is open. The base-collector of a transistor is a diode. So, the collector-base voltage is the maximum reverse voltage that can be applied across to this diode. Take note not to exceed this value. Otherwise, the transistor will get damage right away.

Absolute Maximum Ratings (@ $T_A = +25^\circ\text{C}$, unless otherwise specified.)			
Characteristic	Symbol	Value	Unit
Collector-Base Voltage	V_{CBO}	70	V
Collector-Emitter Voltage	V_{CEQ}	50	V
Emitter-Base Voltage	V_{EBQ}	7	V
Continuous Collector Current	I_C	2	A
Peak Pulse Collector Current	I_{CM}	3	A

6. Saturation Voltage

Another parameter that is important is the saturation voltage. The collector – emitter saturation voltage is needed in order to compute the actual power dissipation of transistor. The ideal case is that this power dissipation is low. In order to attain it, the collector – emitter saturation voltage must be very low.

Collector-Emitter Saturation Voltage (Note 9)	$V_{CE(sat)}$	—	—	0.3	V	$I_C = 1\text{A}, I_B = 50\text{mA}$
Base-Emitter Saturation Voltage (Note 9)	$V_{BE(sat)}$	—	—	1.2	V	$I_C = 1\text{A}, I_B = 50\text{mA}$

The base – emitter saturation voltage is another important thing in selecting a transistor. This is still related to the actual power dissipation. The lower this value, the lower the power dissipation due to the base current. The base – emitter saturation voltage is also an indicative value that needs to be satisfied by the circuit bias. Otherwise, the transistor will not turn on properly.

7. Power Dissipation

The next very important rating of a transistor is the power dissipation. It is given in the datasheet like below.

Thermal Characteristics (@T _A = +25°C, unless otherwise specified.)				
Characteristic		Symbol	Value	Unit
Power Dissipation	(Note 5)	P _D	2.60	W
	(Note 6)		2.30	
	(Note 7)		1.45	
Thermal Resistance, Junction to Ambient Air	(Note 5)	R _{θJA}	48	°C/W
	(Note 6)		54	
	(Note 7)		86	
Operating and Storage Temperature Range		T _J , T _{STG}	-55 to +150	°C

However, the provided value in the table is taken at nominal temperature which is usually at 25°C. This means, when the operating temperature is not anymore 25°C, the values in the table above is no longer valid. In the table above, the power dissipation is specified in Note 5, 6 and 7.

Notes:

5. For a device mounted with the exposed collector pad on 25mm x 25mm 2oz copper that is on a single-sided 1.6mm FR4 PCB; device is measured under still air conditions whilst operating in a steady-state.
6. Same as note (5), except mounted on 25mm x 25mm 1oz copper.
7. Same as note (5), except mounted on minimum recommended pad (MRP) layout.
8. Refer to JEDEC specification JESD22-A114 and JESD22-A115.

Compute the actual power dissipation of the circuit and do not let it exceeds the values specified in the table above. A 50% limit is a good value. For instance at Note 5, the power dissipation rating is 2.6W, thus limit the actual power dissipation to 1.3W.

8. Thermal Resistance

When the transistor is use to operate at temperature more than the typical value, thermal resistance is needed to get the maximum power rating of the transistor. This is also called the de-rated power. Thermal resistance is could be defined as junction to ambient or junction to case. In below table for instance, it is junction to ambient.

Thermal Characteristics (@T _A = +25°C, unless otherwise specified.)				
Characteristic		Symbol	Value	Unit
Power Dissipation	(Note 5)	P _D	2.60	W
	(Note 6)		2.30	
	(Note 7)		1.45	
Thermal Resistance, Junction to Ambient Air	(Note 5)	R _{θJA}	48	°C/W
	(Note 6)		54	
	(Note 7)		86	
Operating and Storage Temperature Range		T _J , T _{STG}	-55 to +150	°C

For instance the actual temperature of operation is 50°C; the power dissipation of the transistor is not anymore the same as specified in the table above. Continue reading below to learn on how to compute the new power dissipation.

9. Operating Temperature

Operating the transistor outside the operating range will lead to immediate failure. In below table, both operating and storage temperatures are provided.

Thermal Characteristics (@T _A = +25°C, unless otherwise specified.)				
Characteristic		Symbol	Value	Unit
Power Dissipation	(Note 5)	P _D	2.60	W
	(Note 6)		2.30	
	(Note 7)		1.45	
Thermal Resistance, Junction to Ambient Air	(Note 5)	R _{θJA}	48	°C/W
	(Note 6)		54	
	(Note 7)		86	
Operating and Storage Temperature Range		T _J , T _{STG}	-55 to +150	°C

Always follow the datasheet. However, it does not mean that you can operate the transistor always at +100°C since it is way lower than maximum limit of +150°C. You need to do little math to check if it is possible. The next topic will discuss this, so keep reading.

How to Compute the De-rated Power Dissipation

The datasheet usually provide the power rating at 25°C like in the table above. This is not useful anymore when the transistor is operating above 25°C.

The de-rated power is can be computed using below equation.

$$\text{Power de-rated} = (T_{j\max} - T_{\text{ambient}}) / R_{\theta ja}$$

T_{jmax} is the maximum junction temperature specified in the datasheet, T_{ambient} on the other hand is the actual ambient temperature of the transistor while R_{θja} is the thermal resistance from junction to ambient. So, using the values provided above; T_{jmax} = +150°C and R_{θja} = 48°C/W at note 5 and the actual operating ambient temperature is 50°C.

$$\text{Power de-rated} = (T_{j\max} - T_{\text{ambient}}) / R_{\theta ja} = (150 - 50) / 48 = \mathbf{2.08 \text{ W}}$$

The resulting power rating or the de-rated power is 2.08W. This is lower than the power dissipation provided in the table above which is 2.6W at Note 5.

Given the actual power dissipation, the maximum junction temperature that the transistor could

operate is can be derived.

$$\text{Actual Power Dissipation} = (T_{j\max} - T_{\text{ambient}}) / R_{\text{thja}}$$

$$T_{j\max} = \text{Actual Power Dissipation} \times R_{\text{thja}} + T_{\text{ambient}}$$

Assuming the actual power dissipation is 3W, the actual maximum junction temperature at an ambient temperature of 50°C is

$$T_{j\max} = \text{Actual Power Dissipation} \times R_{\text{thja}} + T_{\text{ambient}} = 3\text{W} \times 48^\circ\text{C/W} + 50^\circ\text{C} = \mathbf{194^\circ\text{C}}$$

The maximum junction temperature specified in the datasheet is only 150°C, thus the transistor will surely get damaged.

10. Gain and Bandwidth

All considerations above on how to select a transistor are about robustness and reliability related. On the other hand, gain and bandwidth are both needed when the transistor is used as an amplifier. Gain or DC current gain, beta of sometimes called HFE is needed to compute the collector current. From the term gain, it will amplify a small base current and convert to a big collector current.

The bandwidth is also important when operating as an amplifier. The amplifier will not give quality audio when the bandwidth is not sufficient.

Electrical Characteristics (@ $T_A = +25^\circ\text{C}$, unless otherwise specified.)

Characteristic	Symbol	Min	Typ	Max	Unit	Test Condition
Collector-Base Breakdown Voltage	BV_{CBO}	70	—	—	V	$I_C = 100\mu\text{A}$
Collector-Emitter Breakdown Voltage (Note 9)	BV_{CEO}	50	—	—	V	$I_C = 10\text{mA}$
Emitter-Base Breakdown Voltage	BV_{EBO}	7	—	—	V	$I_E = 100\mu\text{A}$
Collector Cut-off Current	I_{CES}	—	—	100	nA	$V_{CE} = 50\text{V}$
Collector-Base Cut-off Current	I_{CBO}	—	—	100	nA	$V_{CB} = 70\text{V}$
Emitter Cut-off Current	I_{EBO}	—	—	100	nA	$V_{EB} = 6\text{V}$
Collector-Emitter Saturation Voltage (Note 9)	$V_{CE(sat)}$	—	—	0.3	V	$I_C = 1\text{A}, I_B = 50\text{mA}$
Base-Emitter Saturation Voltage (Note 9)	$V_{BE(sat)}$	—	—	1.2	V	$I_C = 1\text{A}, I_B = 50\text{mA}$
Base-Emitter Turn-On Voltage (Note 9)	$V_{BE(on)}$	—	—	1.2 0.95	V	$I_C = 1\text{A}, V_{CE} = 2\text{V}$ $I_C = 0.75\text{A}, V_{CE} = 1.6\text{V}$
DC Current Gain (Note 9)	h_{FE}	120 40 80	— — —	— — —	—	$V_{CE} = 2\text{V}, I_C = 0.5\text{A}$ $V_{CE} = 2\text{V}, I_C = 2\text{A}$ $V_{CE} = 1.6\text{V}, I_C = 0.75\text{A}$
Current Gain-Bandwidth Product	f_T	65	—	—	MHz	$I_C = 0.1\text{A}, V_{CE} = 10\text{V}, f = 100\text{MHz}$

11. Dynamic Parameters

When the transistor is used as a high speed switch like those in switching converters, power supplies, inverters and the likes, the dynamic parameters like output capacitance, input capacitance, delay and rise time and fall time is important. Imagine how fast the transistor is switching on and off and the rise and fall times are long, this will result to a very high switching losses. Switching losses is just another form of power dissipation but due to the continuous switching of the transistor.

Electrical Characteristics (@T_A = +25°C, unless otherwise specified.)

Characteristic	Symbol	Min	Typ	Max	Unit	Test Condition
Collector-Base Breakdown Voltage	BV _{CBO}	70	—	—	V	I _C = 100μA
Collector-Emitter Breakdown Voltage (Note 9)	BV _{CEO}	50	—	—	V	I _C = 10mA
Emitter-Base Breakdown Voltage	BV _{EBO}	7	—	—	V	I _E = 100μA
Collector Cut-off Current	I _{CES}	—	—	100	nA	V _{CE} = 50V
Collector-Base Cut-off Current	I _{CBO}	—	—	100	nA	V _{CB} = 70V
Emitter Cut-off Current	I _{EBO}	—	—	100	nA	V _{EB} = 6V
Collector-Emitter Saturation Voltage (Note 9)	V _{CE(sat)}	—	—	0.3	V	I _C = 1A, I _B = 50mA
Base-Emitter Saturation Voltage (Note 9)	V _{BE(sat)}	—	—	1.2	V	I _C = 1A, I _B = 50mA
Base-Emitter Turn-On Voltage (Note 9)	V _{BE(on)}	—	—	1.2 0.95	V	I _C = 1A, V _{CE} = 2V I _C = 0.75A, V _{CE} = 1.6V
DC Current Gain (Note 9)	h _{FE}	120 40 80	—	—	—	V _{CE} = 2V, I _C = 0.5A V _{CE} = 2V, I _C = 2A V _{CE} = 1.6V, I _C = 0.75A
Current Gain-Bandwidth Product	f _T	65	—	—	MHz	I _C = 0.1A, V _{CE} = 10V, f = 100MHz
Output Capacitance	C _{obo}	—	26	—	pF	V _{CB} = 10V, f = 1MHz
Input Capacitance	C _{ibo}	—	45	—	pF	V _{EB} = 0.5V, f = 1MHz
Delay Time	t _d	—	29	—	ns	I _C = 0.5A, V _{CC} = 10V I _{B1} = -I _{B2} = 50mA
Rise Time	t _r	—	20	—	ns	
Storage Time	t _s	—	378	—	ns	
Fall Time	t _f	—	57	—	ns	

12. Transistor Type, NPN or PNP

The last thing to take note on how to select a transistor is the type. You must know if you need a NPN or PNP type. These two are totally different. NPN is best to use for low side driving and switches while PNP is best to use as high side driving and switch. The electrical analysis is common for both type however if you are not usually used the PNP, it is little confusing.

Should you have inputs, feel free to comment in the comment section. We will try to answer your questions. We are also offering a consultation.

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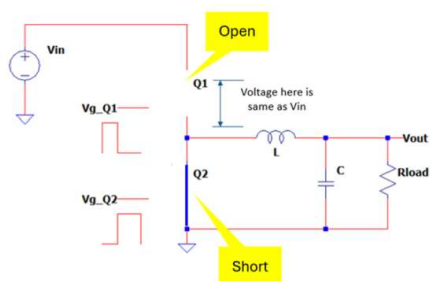
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Quick Facts About Synchronous Buck Converter

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SUPPLY VOLTAGE		
Parameters	Value	Unit
V _{DS}	5	volt
V _{DO}	5	volt
This is the typical value of V _{DS} . See the figure. This is the typical value of V _{DO} . See the figure.		
AMBIENT TEMPERATURE		
Parameters	Value	Unit
T _{amb} max	60	°C
T _{amb} min	-30	°C
This is the maximum operating ambient temperature. This is a design given. This is the minimum operating ambient temperature. This is a design given.		
RESISTOR DATA		
Parameters	Value	Unit
R _{G1}	1500	ohms
R _{G2}	100000	ohms
P _R R _{G1} rating	0.1	watt
P _R R _{G2} rating	0.1	watt
Input here the resistance of R _{G1} . See Figure. Input here the resistance of R _{G2} . See Figure. Input here the power dissipation rating of R _{G1} . Get this from the datasheet. Input here the power dissipation rating of R _{G2} . Get this from the datasheet.		
RELAY DATA		
Parameters	Value	Unit
R _{COIL} typical	100	ohms
L _{COIL}	100	uH
P _{COIL} rating	1	watt
T _{min} relay	-40	°C
T _{max} relay	125	°C
Input here the coil resistance of the relay. Get this from the datasheet. Input here the coil inductance of the relay. Get this from the datasheet. Input here the power dissipation rating of the relay coil. Get this from the datasheet. Input here the minimum operating temperature of the relay. Get this from the datasheet. Input here the maximum operating temperature of the relay. Get this from the datasheet.		

Design Template for NMOS Lowside Relay Driver

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Buck Converter Design Using Excel Calculator

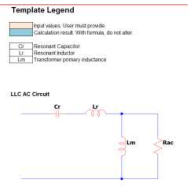
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Synchronous Buck Converter Design Calculator Excel-External Switches

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Parameter	Symbol	Value	Unit
Input Voltage			
Maximum input voltage	V _{in} max	100	V
Minimum input voltage	V _{in} min	20	V
Rated input voltage	V _{in} rated	40	V
Output Voltage			
Maximum output voltage	V _o max	15	V
Minimum output voltage	V _o min	1.5	V
Rated output voltage	V _o rated	1.5	V
Load Current			
Maximum output current (load full)	I _o max	24	A
Minimum output current (load full)	I _o min	24	A
Rated output current (load full)	I _o rated	24	A
Frequency Range			
Maximum frequency	f _{max}	100	Hz
Minimum frequency	f _{min}	400	Hz
Rated frequency	f _{rated}	400	Hz
Inductor Value Range			
Maximum inductor value	L _{max}	100	uH
Minimum inductor value	L _{min}	100	uH
Rated inductor value	L _{rated}	100	uH
Capacitor Value Range			
Maximum capacitor value	C _{max}	100	uF
Minimum capacitor value	C _{min}	10	uF
Rated capacitor value	C _{rated}	10	uF
Maximum gate	G _{max}	1.5	uS
Minimum gate	G _{min}	1.5	uS
Rated gate	G _{rated}	1.5	uS



Half Bridge LLC Resonant Converter Power Section Calculator

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